

QUANT RESEARCH TRAINING + THESIS EXECUTION PLAN

A 12-week roadmap for building an industry-style ML pipeline while developing an academically defensible thesis on IV surfaces, bubbles, and return predictability

Primary career goal	Quantitative research / trading role in Taiwan
Immediate skill goal	End-to-end cross-sectional ML alpha research pipeline
Thesis goal	Explain why option-implied information / IV surfaces predict future returns, with bubble-related mechanisms and competing explanations
Operating rule	Use ML as a research tool; do not make "best model wins" the thesis contribution

Prepared from your current CV, your 2026 Taiwan quantitative-trading recruitment workshop notes, the Kelly IV-surface research workflow, and the Jarrow-Kwok bubble paper.

1. The strategy in one page

Your two-track plan

Track A (next 3 weeks): build a complete ML alpha-research project to learn the workflow you want for recruiting.

Track B (weeks 4-12 and beyond): turn the thesis into an economic-mechanism study: why option-implied information predicts returns, whether bubble information helps explain it, and whether the relationship survives conventional controls.

Why this split fits your profile. Your CV already demonstrates time-series forecasting, IV surfaces, VAR/MAR, LSTM/PyTorch, and expanding-window validation. The missing evidence is more cross-sectional: raw feature engineering, tabular ML, signal ranking, portfolio formation, transaction costs, factor attribution, and model interpretation. The recruiting workshop you attended explicitly emphasized data quality, factor discovery, backtesting, tradability, overfitting control, and the fact that tree-based models can be common in practice.

What you are not doing. You are not replacing the thesis with an XGBoost benchmark. You are also not forcing a Kelly replication when the necessary 100-stock IV-surface panel is unavailable. You are borrowing Kelly's research workflow and using a separate, accessible dataset to learn it.

North-star thesis question

Why can information embedded in the implied-volatility surface forecast subsequent returns, and to what extent is that predictability associated with bubble/mispricing information rather than conventional price, liquidity, risk, fundamental, or macro predictors?

12-week milestone map

Weeks	Primary focus	Main output	Decision gate
1-3	Quant ML alpha project	Reproducible research repo + backtest + short report	Can you produce leakage-safe OOS signals and portfolios?
4-5	Thesis theory + data feasibility	Research question, hypotheses, data memo, Jarrow-Kwok replication start	Is the option/bubble dataset sufficient for your exact thesis unit of analysis?
6-7	Baseline econometrics	Linear evidence with economically motivated controls	Does option/bubble information add incremental explanatory/predictive content?
8-9	Nonlinear ML extension	RF / boosting + interpretation	Are nonlinearities/interactions real and stable OOS?
10-11	Economic tests + robustness	Portfolio/factor/regime/ablation evidence	Does the signal survive costs, controls, and alternative specifications?
12	Write + present	Thesis draft core chapters + 10-minute research talk	Can you explain mechanism, limitations, and evidence clearly?

2. Track A - 3-week quant research bootcamp

Purpose: build the full pipeline once, quickly, on data you can actually obtain. The project does not need academic novelty. Its purpose is skill acquisition and interview evidence.

Project research question

Can a panel of observable equity characteristics produce stable out-of-sample cross-sectional return rankings, and how do linear, tree-based, and neural models differ in predictive behavior, turnover, and economic value?

2.1 Recommended project design

Unit of observation	Stock i at month t . One row = one stock-month.
Target	Next-month total return $R(i,t+1)$, plus next-month cross-sectional return rank as a secondary target/diagnostic.
Universe	Start with 100-300 liquid equities with clean price/volume history. Taiwan equities are attractive for local-market relevance; U.S. equities are also acceptable if cleaner and faster to obtain.
Frequency	Monthly prediction/rebalancing first. Do not start intraday.
Feature timing	Every feature must use information available by the prediction date. Accounting variables require reporting-date lags; price/volume features are easier to make point-in-time.
Validation	Chronological expanding or rolling windows. Never random train/test splits across time.
Portfolio	Rank predicted returns each month; evaluate top-bottom quantile spread and long-only top-quantile variant.
Primary success criterion	A correct, reproducible, leakage-resistant pipeline - not a spectacular Sharpe ratio.

2.2 Feature library to learn

Implement a small, disciplined library rather than 200 signals. Aim for roughly 20-40 features across these families:

Family	Examples / learning objective
Price / technical	1, 3, 6, 12-month momentum; short-term reversal; moving-average distance; realized volatility; downside volatility; max drawdown.
Trading / liquidity / microstructure	Volume, turnover, Amihud illiquidity, zero-return frequency, bid-ask spread if available, volume shock.

Risk exposures	Market beta, idiosyncratic volatility, downside beta, residual momentum if you implement a factor regression.
Fundamental / accounting	Size, book-to-market, profitability, investment, leverage, earnings yield - only if you can obtain properly lagged historical values.
Macro / market state	Market return, market volatility proxy, rates or yield-curve variables - use as regime/context variables rather than pretending they are firm-level characteristics.
Options-derived (optional)	Do not block the project on options. Add simple option-derived variables later only if accessible.

2.3 Model ladder

Naive baselines: Historical mean; zero-return prediction; simple momentum sort.

Linear: OLS for interpretation, Ridge, Lasso, Elastic Net.

Bagged trees: Random Forest or ExtraTrees.

Boosted trees: Choose one primary library first: XGBoost or LightGBM. Add CatBoost only if there is a concrete reason.

Neural benchmark: Small MLP with strong regularization. This is a benchmark, not automatically the final model.

Rule for model selection

Do not optimize the final test Sharpe. Hyperparameters are selected only using training/validation periods. The test period exists to estimate generalization, not to make research decisions.

3. Week 1 - Data, targets, and feature engineering

End-of-week deliverable: a single clean stock-month panel and a feature-generation pipeline that can be rerun from raw data.

Day 1 - Freeze the experiment contract

- ☐ Write a one-page README: universe, dates, rebalance frequency, target definition, prediction timestamp, and data sources.
- ☐ Decide the primary target: next-month return. Do not change it every two days.
- ☐ Define an initial OOS period before looking at model performance.
- ☐ Create a data dictionary with every raw field and its timestamp meaning.
- ☐ Create a Git repository structure before analysis begins.

Day 2 - Build raw-to-panel data

- ☐ Ingest adjusted prices and volume; calculate month-end observations.
- ☐ Create next-month returns by shifting the target forward within each stock.
- ☐ Drop rows only for explicit reasons and log the counts.
- ☐ Check duplicates, split adjustments, stale prices, missing months, and survivorship issues.
- ☐ Save an immutable intermediate panel (for example Parquet) so feature code does not repeatedly hit external data sources.

Day 3 - Price, volatility, and liquidity features

- ☐ Implement momentum and reversal features using lagged returns.
- ☐ Implement realized volatility and downside volatility.
- ☐ Implement turnover / volume shock / Amihud-style illiquidity if inputs exist.
- ☐ Unit-test at least five hand-calculated stock-month examples.
- ☐ Plot feature distributions through time; look for scale breaks and impossible values.

Day 4 - Cross-sectional preprocessing

- ☐ Winsorize or clip extreme features using rules fit only on the available historical window.
- ☐ Cross-sectionally rank or z-score features by month as a second representation.
- ☐ Keep missingness indicators where economically meaningful; do not silently fill everything with zero.
- ☐ Build one sklearn-compatible preprocessing pipeline to avoid train/test inconsistencies.

Day 5 - Exploratory research memo

- ☐ Compute feature coverage by year and stock.
- ☐ Compute pairwise correlations and a PCA diagnostic. PCA is exploratory here, not a mandatory alpha model.
- ☐ Check monotonic return sorts for a few canonical features such as momentum and volatility.
- ☐ Write one page: what looks reasonable, what looks suspicious, and what you will not fix by data snooping.

Week 1 stop rule

If the panel is not trustworthy by Friday, do not move to XGBoost. Fix timestamps and features first. Most bad quant projects are data problems disguised as modeling problems.

4. Week 2 - Models and leakage-safe validation

End-of-week deliverable: a common evaluation harness comparing linear, tree-based, and neural baselines on identical OOS months.

Walk-forward protocol

Use the same protocol for every model:

1. Choose an initial training window.
2. Within the historical training period, create a validation scheme for hyperparameter selection.
3. Fit on the allowed past data only.
4. Predict the next unseen month (or block of months).
5. Advance the window and repeat.
6. Store every OOS prediction with model name, training cutoff, validation choice, and timestamp.

Model implementation order

- ☐ OLS / Ridge / Elastic Net: establish a transparent benchmark and coefficient stability.
- ☐ Random Forest: learn bagging, tree depth, minimum leaf size, feature subsampling, and permutation importance.
- ☐ XGBoost or LightGBM: learn boosting, learning rate, depth/leaves, subsampling, regularization, and early stopping.
- ☐ Small MLP: reuse your PyTorch experience but keep architecture intentionally simple.
- ☐ Optional: compare raw features vs monthly cross-sectional ranks.

Metrics to store for every model

Category	Metrics	Why it matters
Statistical	MSE / MAE; OOS R-squared	Baseline forecast quality; often weak in returns, so interpret modest values carefully.
Cross-sectional	Spearman rank IC; mean IC; IC t-stat; top-bottom spread in realized returns	Closer to the stock-ranking use case.
Portfolio	Annualized return/volatility; Sharpe; max drawdown; turnover	Economic value and risk.
Stability	Performance by year/regime; feature importance stability	Detects one-period luck and regime dependence.

What counts as progress

A tree model beating Ridge by a tiny amount is not the main achievement. The achievement is that the comparison is fair, OOS, reproducible, and tied to a trading decision.

5. Week 3 - Portfolio, factor attribution, and interpretation

End-of-week deliverable: a polished project that looks like quant research rather than a Kaggle notebook.

Portfolio construction

- ☐ Rank OOS predictions each month.
- ☐ Form quintile portfolios first; use deciles only if the cross-section is large enough.
- ☐ Report equal-weight long-short and long-only top-quantile variants.
- ☐ Lag portfolio implementation consistently so prediction timestamps and execution timestamps are realistic.
- ☐ Calculate monthly turnover and apply at least two transaction-cost assumptions.
- ☐ Plot cumulative returns, drawdowns, and rolling Sharpe / rolling IC.

Factor attribution

Regress portfolio returns on an appropriate factor set for the market you use. The purpose is not to "prove alpha"; it is to ask whether your strategy is mainly a disguised exposure to familiar risks such as market, size, value, or momentum.

- ☐ Report alpha, factor loadings, standard errors / t-statistics, and sample period.
- ☐ Compare raw long-short returns with factor-adjusted results.
- ☐ Check whether model performance collapses after neutralizing obvious exposures.

Interpretation and ablation

- ☐ Permutation importance for all models where sensible.
- ☐ SHAP for the boosted-tree model, with both global importance and a small number of interaction plots.
- ☐ Feature-family ablations: price only; price + liquidity; + risk; + fundamentals; + macro.
- ☐ Rerun without the single strongest feature. If the strategy dies, say so.
- ☐ Compare feature importance across subperiods rather than trusting one full-sample plot.

Final repository layout

```
quant-alpha-pipeline/  
  README.md  
  config/  
  data/README.md          # no proprietary raw data  
  src/data.py  
  src/features.py  
  src/validation.py  
  src/models.py  
  src/portfolio.py  
  src/factors.py  
  src/interpret.py  
  notebooks/01_data_audit.ipynb  
  notebooks/02_features.ipynb  
  notebooks/03_models.ipynb  
  notebooks/04_backtest.ipynb  
  notebooks/05_interpretation.ipynb  
  reports/research_note.pdf  
  tests/
```

Project completion definition

- ☐ One-command or clearly documented reproduction from cached panel to final tables.
- ☐ At least one linear model, one tree ensemble, one boosted-tree model, and one simple neural benchmark.
- ☐ All headline results are genuinely OOS.
- ☐ Backtest includes turnover and transaction costs.
- ☐ At least one factor attribution table.
- ☐ At least one feature-family ablation and one regime robustness test.

□ A 2-4 page research note that includes negative results and limitations.

Then stop

After Week 3, freeze this project. Do not spend another month squeezing 5 basis points of Sharpe from it. Move the tools you learned into the thesis.

6. Track B - Thesis: IV surfaces, bubbles, and return predictability

Core principle: the thesis is an economic explanation problem, not a model horse race.

6.1 Working thesis framing

Working question: Why does information in the implied-volatility surface predict subsequent returns? Is part of that relation associated with bubble/mispricing information, and does it survive conventional predictors of returns and risk?

6.2 Hypothesis ladder

Hypothesis	Interpretation
H1 - Predictive content	Option-implied / IV-surface information at t is associated with subsequent returns.
H2 - Bubble link	The IV-surface signal is related to a theoretically motivated option-based bubble measure or bubble state.
H3 - Mechanism relevance	Adding the bubble measure changes the magnitude, significance, or economic behavior of the IV-return relationship.
H4 - Incremental information	The relationship is not fully subsumed by momentum, realized volatility, liquidity, risk exposures, fundamentals, or market-state variables.
H5 - Nonlinearity (extension)	Interactions among IV/bubble information and conventional variables are economically meaningful and stable OOS.

6.3 What each feature family is for

Do not throw every variable into XGBoost without a reason. Give each group a role in the research design:

IV / options: Main information set whose predictive content you are trying to explain.

Bubble measure: Candidate mechanism suggested by your professor and Jarrow-Kwok.

Price / technical: Tests whether the IV signal is just momentum, reversal, or recent volatility in disguise.

Liquidity / microstructure: Tests whether limits to arbitrage, trading frictions, or liquidity conditions help explain the signal.

Risk / factors: Tests whether return predictability is compensation for conventional systematic exposures.

Fundamentals: Tests whether option signals proxy for valuation, profitability, leverage, investment, or earnings information.

Macro / regime: Tests whether the relation is stronger in high-volatility, stress, rate, or market-state regimes.

Critical academic distinction

Tree-based models can reveal nonlinear predictive associations and interactions. They do not by themselves establish a causal explanation. Use ML to discover and characterize patterns; use theory and econometric tests to argue about mechanism.

7. Weeks 4-5 - Theory, replication, and the data feasibility gate

Goal: decide the exact unit of analysis before building the thesis model.

7.1 Read Jarrow-Kwok as a mechanism paper

- ☐ Reproduce the paper's definition of a bubble and write it in your own notation.
- ☐ Understand why option prices can partially identify a bubble under no-arbitrage and data constraints.
- ☐ Distinguish what is theoretically identified from what is estimated empirically.
- ☐ Reproduce one central table/figure or a minimal version using the authors' public replication data before modifying the method.
- ☐ Write a two-page "mechanism memo": assumptions, identifying logic, predicted sign/direction, and limitations.

Verified source: Jarrow, R. A. & Kwok, S. S. (2021), "Inferring financial bubbles from option data," Journal of Applied Econometrics 36(7), 1013-1046. The publisher marks the article with an Open Data badge and links the JAE replication archive: <http://qed.econ.queensu.ca/jae/datasets/jarrow001/>.

7.2 Data feasibility decision

Data situation	What you can study	Suitable methods	Do not do
A. Individual-stock IV panel becomes available	Cross-sectional stock-return prediction + bubble/controls	Panel regressions, portfolio sorts, boosting, SHAP, factor attribution	Do not change vendors or surface definitions midstream without documenting it.
B. Only SPX IV surface + Jarrow-Kwok index data	Market/index return predictability + bubble mechanism	Low-dimensional econometrics; carefully regularized ML only as robustness	Do not train a high-capacity model on ~120 monthly observations and call it evidence.
C. Bubble data are reproducible but stock IV panel remains inaccessible	Bubble/option-price mechanism at index level; conventional predictors as controls	Replication + extension + regime/nonlinearity analysis	Do not force a fake Kelly cross-section from one SPX surface.
D. Better option dataset appears later	Add as extension after core thesis works	Repeat pre-specified tests	Do not delay the thesis indefinitely while searching for the perfect dataset.

Week 5 decision deadline

At the end of Week 5, freeze the thesis dataset and unit of analysis. If a broad stock-level IV panel is still not realistically obtainable, formally pivot the empirical thesis to the index-level bubble/option setting rather than spending additional weeks on data hunting.

8. Weeks 6-7 - Baseline econometrics before ML

Build the academic backbone first. The exact regression depends on the final data structure, but the logic should progress in layers.

Layered specification

1. Univariate predictive relation: future return on IV/option signal.
2. Add bubble measure.
3. Add conventional return predictors by economic family.
4. Add risk/factor controls and market-state controls.
5. Test economically motivated interactions suggested by theory.
6. Evaluate out-of-sample stability, not only in-sample t-statistics.

Conceptual form: $R(t+1) = \alpha + \beta_{IV} * IVSignal(t) + \beta_B * Bubble(t) + \gamma' X(t) + error(t+1)$.

Required diagnostics

- ☐ Coefficient signs and confidence intervals, not only p-values.
- ☐ Stability across subperiods / regimes.
- ☐ Collinearity between IV, bubble, realized volatility, and market-state variables.
- ☐ Alternative definitions of the main signal.
- ☐ Lag sensitivity and forecast horizon sensitivity.
- ☐ Out-of-sample forecast comparison against a simple historical benchmark.
- ☐ Clear distinction between predictive association and causal mechanism.

If H1 fails

A null result is acceptable if the design is credible. Do not add features or change horizons repeatedly until significance appears. Document pre-specified primary tests and label exploratory tests as exploratory.

9. Weeks 8-9 - Tree-based ML as a thesis extension

Only after the baseline economics are clear should you fit nonlinear models. The thesis question for ML is not "Which algorithm wins?" but "What nonlinearities or interactions are missed by the linear specification, and are they stable enough to matter?"

Recommended model set

- ☐ Ridge / Elastic Net as the regularized linear benchmark.
- ☐ Random Forest as a bagged nonlinear benchmark.
- ☐ One boosted-tree model (XGBoost or LightGBM) as the primary nonlinear model.
- ☐ Optional small MLP only if sample size supports it; do not make deep learning mandatory.

Interpretation workflow

1. Measure OOS predictive change relative to the linear baseline.
2. Use permutation importance and SHAP to identify recurring variables/interactions.
3. Check whether the same interaction appears across folds/subperiods.
4. Translate stable ML patterns back into explicit econometric hypotheses.
5. Test those interactions with transparent regressions / subsample sorts.

Example: if boosting repeatedly indicates that an extreme bubble measure matters mainly when IV skew is extreme and liquidity is low, estimate a transparent interaction specification and compare high/low-liquidity subsamples. The ML model becomes a hypothesis generator, not an oracle.

What not to interpret causally

- ☐ SHAP magnitude is not a causal effect.
- ☐ Feature importance does not tell you whether the variable is a risk premium, mispricing, or information proxy.
- ☐ A profitable backtest does not prove a bubble mechanism.
- ☐ A tree interaction found after repeated searching requires out-of-sample confirmation.

10. Weeks 10-11 - Economic tests, portfolios, and robustness

If you have a cross-sectional stock panel

- ☐ Rank the IV/bubble/ML signal by month.
- ☐ Form top-minus-bottom portfolios and report turnover/costs.
- ☐ Run factor regressions on portfolio returns.
- ☐ Compare raw IV signal, bubble-only signal, combined linear signal, and combined ML signal.
- ☐ Test whether performance is concentrated in illiquid, small, volatile, or high-bubble stocks.

If you have index-level data only

- ☐ Use a time-series trading rule only if it follows from a pre-specified signal and realistic lag.
- ☐ Report drawdown, turnover, costs, and regime performance.
- ☐ Compare against buy-and-hold and simple momentum / volatility timing benchmarks.
- ☐ Use bootstrap or other appropriate uncertainty procedures where the paper/theory motivates them.
- ☐ Keep model complexity low relative to the effective sample size.

Robustness matrix

Dimension	Minimum test	Question answered
Time	Early vs late period; stress vs calm	Is the relation regime-specific?
Features	Remove one family at a time	Which information set actually matters?
Signal definition	Alternative IV/bubble definitions	Is the result definition-sensitive?
Costs	Low / medium / high trading-cost assumptions	Is apparent alpha implementable?
Model	Linear vs RF vs boosting	Is nonlinearity genuinely useful?
Horizon	At least one adjacent horizon	Is the finding a single-horizon artifact?

11. Week 12 - Write the thesis like a researcher

Recommended chapter skeleton

- 1. Introduction:** Economic puzzle, contribution, main findings, limitations.
- 2. Literature and mechanism:** IV-return predictability, option-implied information, bubble theory, competing explanations.
- 3. Data and construction:** Option/IV data, bubble measure, predictors, returns, timing, filters, missingness.
- 4. Baseline empirical design:** Predictive regressions and hypothesis tests.
- 5. Nonlinear ML extension:** Validation, RF/boosting, interpretation, translated interaction tests.
- 6. Economic significance:** Portfolios/trading rule, costs, factor adjustment, regime analysis.
- 7. Robustness and limitations:** Alternative definitions, subsamples, model risk, sample limits.
- 8. Conclusion:** What the evidence says - and what it does not say - about why IV contains return information.

Ten-minute presentation structure

1. The puzzle: IV surfaces predict returns, but what information are they capturing?
2. Mechanism: bubble-related option information and competing explanations.

3. Data and identification / timing.
4. Baseline linear result.
5. What ML adds: one or two stable nonlinear patterns.
6. Economic significance / portfolio or trading implication.
7. Robustness and limitations.
8. Takeaway: what you learned about the mechanism, not which model had the highest score.

12. Research governance - rules that prevent wasted months

Data deadline	No thesis data hunting after the Week 5 decision gate unless a source is already confirmed and can be integrated in <2 days.
Model budget	For the practice project: at most one model per major family. For the thesis: only models tied to a specific research question.
Test-set discipline	No hyperparameter or feature decisions based on final OOS performance.
Feature discipline	Every thesis feature family needs an economic role: main signal, mechanism, control, alternative explanation, or regime variable.
Novelty discipline	Do not claim novelty because you used XGBoost, SHAP, a Transformer, or a CNN.
Interpretation discipline	Predictive association != causality; feature importance != mechanism.
Writing discipline	Write one page of research notes every Friday. Methods/results that are not documented during the project are usually forgotten or rationalized later.
Stopping rule	Once the minimum viable thesis answers H1-H4 credibly, extensions are optional.

13. Skill checklist for 2027 quant recruiting

Use this as a progress tracker. It is tailored to the gaps between your current CV and the skills emphasized in the recruiting workshop.

Skill	Evidence to produce	Where	Done
Data / research engineering	Point-in-time panel; missingness audit; reproducible feature pipeline; unit tests	Track A Week 1	☐
Linear/statistical ML	Ridge/Lasso/Elastic Net; scaling; regularization; coefficient stability	Track A Week 2	☐

Tree ensembles	RF; XGBoost/LightGBM; early stopping; regularization; importance	Track A Week 2	☐
Validation	Expanding/rolling CV; nested tuning; leakage checks	Track A Week 2	☐
Cross-sectional research	IC; rank sorts; signal decay; quantile portfolios	Track A Weeks 2-3	☐
Backtesting	Execution lag; turnover; costs; drawdown; capacity awareness	Track A Week 3	☐
Asset-pricing diagnostics	Factor regressions; alpha vs exposure; risk decomposition	Track A Week 3 / thesis	☐
Interpretability	Permutation importance; SHAP; interaction stability; ablations	Track A Week 3 / thesis	☐
Research judgment	Hypothesis before model; robustness; negative results; limitations	Every week	☐
Communication	2-4 page research note; 10-minute talk; concise README	Week 3 and Week 12	☐

14. Minimum viable thesis vs optional extensions

Minimum viable thesis - must finish	Optional extensions - only after core works
☐ Theory-grounded bubble / option measure	☐ CNN / image representation if a real stock IV panel becomes available
☐ Clear IV/option signal	☐ Deep multimodal sentiment model
☐ Baseline predictive regression	☐ Additional tree libraries beyond the first boosted-tree implementation
☐ Conventional control families	☐ PCA / principal-feature decomposition beyond basic diagnostics
☐ One leakage-safe OOS evaluation	☐ International replication
☐ One nonlinear tree-model extension	☐ Intraday execution modeling
☐ Robustness across time / specification	☐ Alternative option-vendor comparison
☐ Honest limitations and data discussion	

15. Weekly operating system

Use the same rhythm for both the practice project and thesis so you build research habits rather than only code.

Monday: Define one question and one measurable deliverable for the week.

Tuesday: Implement / collect data. Add tests and audit plots.

Wednesday: Run primary experiments. Do not branch into five new ideas.

Thursday: Robustness, ablations, and failure analysis.

Friday: Write a one-page memo: question, method, result, failure modes, next decision. Commit clean code.

Weekend (optional): Read theory/papers or refactor; do not use the weekend to compensate for uncontrolled scope.

Friday memo template

Question -> Data/time alignment -> Method -> Main result -> Robustness -> What failed -> What I believe now -> Next week's single decision.

16. Your next 72 hours

- ☐ Create the practice-project repository and write the one-page experiment contract.
- ☐ Choose a simple, obtainable equity universe and monthly date range. Do not use options for this project.
- ☐ Build raw adjusted price/volume -> month-end stock-month panel -> next-month return target.
- ☐ Implement five features only: 12-month momentum, 1-month reversal, realized volatility, turnover/volume, and market beta.
- ☐ Run a naive momentum sort and Ridge baseline end-to-end before adding Random Forest or XGBoost.
- ☐ Download / inspect the Jarrow-Kwok replication archive separately. Do not mix thesis data work into the practice project yet.
- ☐ Create a thesis-notes document with sections: theory, data, hypotheses, controls, unresolved questions.

By the end of the first 72 hours, you should have one boring but complete pipeline. That is more valuable than a half-finished Transformer, an unverified options dataset, or another week of searching for the perfect thesis.

17. Sources and anchors

Career / recruiting anchor. Your 2026 Quantitative Trading Recruitment Briefing report: emphasizes factor discovery, feature/data quality, backtesting, execution/tradability, risk controls, overfitting control, ownership, and notes that tree-based models may be common in practice rather than neural networks being the default.

Current profile anchor. Your CV: strong GPA and quantitative coursework; projects already include IV-surface forecasting, VAR/MAR, ElasticNet, LSTM/PyTorch, and expanding-window validation. The roadmap therefore emphasizes cross-sectional tabular ML, portfolio construction, factor attribution, and research communication as complementary evidence.

Research workflow anchor. Kelly et al., "Deep Learning from Implied Volatility Surfaces": the useful template for your practice is the end-to-end research workflow - OOS prediction, portfolio sorting, factor analysis, and interpretation - not necessarily the exact OptionMetrics dataset or CNN architecture.

Thesis mechanism anchor. Jarrow, R. A. & Kwok, S. S. (2021), "Inferring financial bubbles from option data," Journal of Applied Econometrics 36(7), 1013-1046, DOI 10.1002/jae.2862. The JAE replication archive is publicly linked by the publisher.

Final instruction to yourself

Do the practice project to become faster and more competent. Do the thesis to answer a real economic question. Let the same tools serve both goals, but do not confuse the tool with the contribution.